Economics?
Economics is the study of how individuals, firms, and societies make choices when faced with scarce resources and unlimited wants. It examines how people allocate resources, how goods and services are produced and distributed, and how these decisions affect overall welfare.

In simple terms:
Economics = The science of choice under scarcity.

Economics is the social science that studies the choices that individuals, businesses, governments, and entire societies make as they cope with scarcity and the incentives that influence and reconcile those choices.

The subject has two parts: ■ Microeconomics ■ Macroeconomics

Microeconomics is the study of the choices that individuals and businesses make, the way these choices interact in markets, and the influence of governments.

Macroeconomics is the study of the performance of the national economy and the global economy.

Example: A farmer deciding how much rice to plant is microeconomics. The government deciding how much to spend on rural development is macroeconomics.

Central Concepts of Economics
a) Scarcity

- Scarcity means that resources (land, labor, capital, entrepreneurship) are limited compared to human wants, which are unlimited.
- Scarcity means there is **not enough of everything for everyone**.
- Because of scarcity, every society must make choices about what to produce, how to produce, and for whom to produce.

Example: If a country has limited steel, it must choose between producing cars, bridges, or railways. Land in Bangladesh is scarce. Should it be used for rice, jute, or industrial development? This reflects the core problem of scarcity and choice. During COVID-19, governments faced tough choices: should factories make ventilators or consumer goods?

Scarcity forces individuals and societies to prioritize and make **trade-offs.**

b) Efficiency

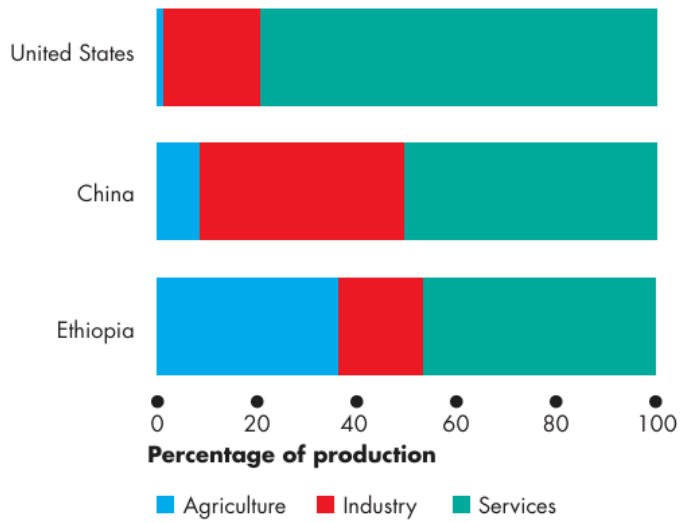
- Efficiency is about making the best possible use of scarce resources.
- Productive efficiency: Producing goods at the lowest possible cost. No resources are wasted; production is at the lowest possible cost.
- Allocative efficiency: Producing the right mix of goods and services that society most desires. Resources are allocated to produce goods and services that provide the greatest benefit to society.

The Problems of Economic Organization

Every economic system faces three fundamental problems:

1. What to produce? – Which goods and services should be produced, and in what quantity? A society must decide whether to use resources for consumer goods (like food, clothes) or capital goods (like machines, factories).

FIGURE 1.1 What Three Countries Produce



Agriculture and manufacturing are small percentages of production in rich countries such as the United States and large percentages of production in poor countries such as Ethiopia. Most of what is produced in the United States is services.

Source of data: CIA Factbook 2017, Central Intelligence Agency.

2. How to produce? – Which techniques and resources should be used? Should production rely more on labor (labor-intensive) or machines (capital-intensive)?

How we produce is described by the technologies and resources that we use. The resources used to produce goods and services are called ***factors of production*** **,** which are grouped into four categories:
■ ***Land*** ■ ***Labor*** ■ ***Capital*** ■ ***Entrepreneurship***

Land The "gifts of nature" that we use to produce goods and services are called land. It includes land in the everyday sense together with minerals, oil, gas, coal, water, air, forests, and fish.
Labor The work time and work effort that people devote to producing goods and services is called labor. Labor includes the physical and mental efforts of all the people who work on farms and construction sites and in factories, shops, and offices.
Capital The tools, instruments, machines, buildings, and other constructions that businesses use to produce goods and services are called capital.
Entrepreneurship The human resource that organizes labor, land, and capital is called entrepreneurship. Entrepreneurs are the drivers of economic progress. They develop new ideas about what and how to produce, make business decisions, and bear the risks that arise from these decisions.

People earn their incomes by selling the services of the factors of production they own: ■ Land earns rent. ■ Labor earns wages. ■ Capital earns interest. ■ Entrepreneurship earns profit.

3. For whom to produce? – How will goods and services be distributed among individuals and groups? How will the goods and services be distributed equally among people, or based on income and purchasing power?

Types of Economic Systems

a) **Market Economy**

- Decisions are made by individuals and firms through interaction in markets. Decentralized decision-making. Households and firms interact in markets to decide production and consumption.
- Prices act as signals of scarcity and incentives for production. Advantage: Encourages innovation and efficiency.
  Disadvantage: Can lead to inequality and market failures.
- Example: USA.

b) **Command Economy**

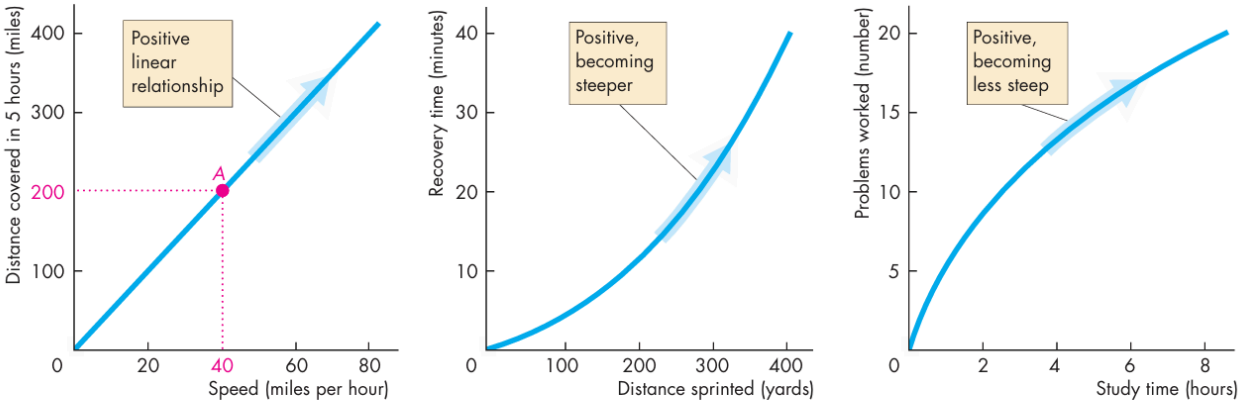
- Decisions are made by the government or central authority. Centralized decision-making. The government decides what to produce, how to produce, and who receives goods.
  Advantage: Can mobilize resources quickly.
  Disadvantage: Often inefficient and unresponsive to consumer needs.
- Resources are allocated according to government plans, not market signals.
- Example: North Korea ,USSR.

c) **Mixed Economy**

- A combination of market forces and government intervention.
- Most countries today (including Bangladesh) have mixed economies.
- Bangladesh – mostly market-based, but the government intervenes in key sectors (subsidies for fertilizer, regulation of rice prices, etc.).

Graphs Used in Economics

FIGURE A1.5 Positive (Direct) Relationships



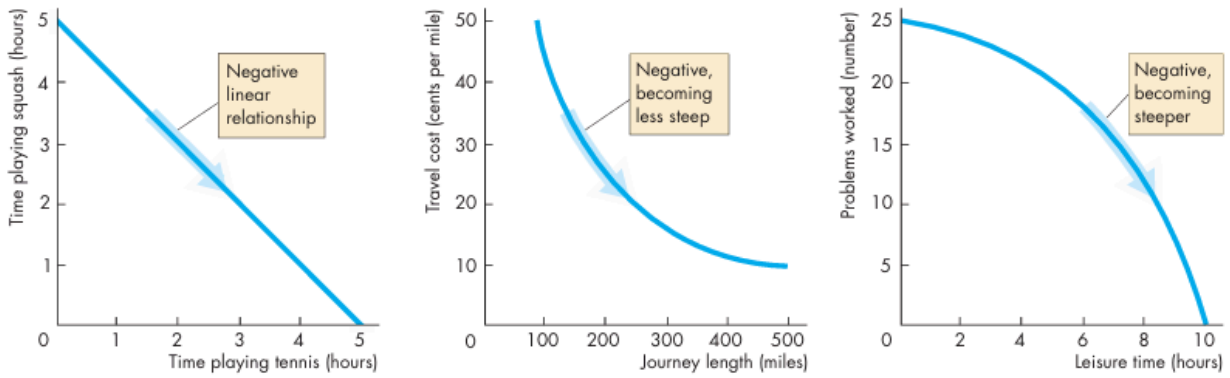
(a) Positive linear relationship **(b) Positive, becoming steeper** **(c) Positive, becoming less steep**

Each part shows a positive (direct) relationship between two variables. That is, as the value of the variable measured on the *x*-axis increases, so does the value of the variable measured on the *y*-axis. Part (a) shows a linear positive relationship—as the two variables increase together, we move along a straight line.

Part (b) shows a positive relationship such that as the two variables increase together, we move along a curve that becomes steeper.

Part (c) shows a positive relationship such that as the two variables increase together, we move along a curve that becomes flatter.

FIGURE A1.6 Negative (Inverse) Relationships



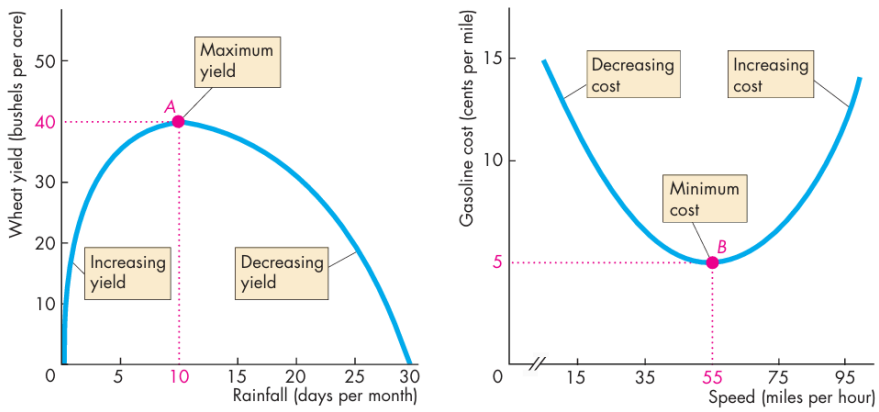
(a) Negative linear relationship **(b) Negative, becoming less steep** **(c) Negative, becoming steeper**

Each part shows a negative (inverse) relationship between two variables. Part (a) shows a linear negative relationship. The total time spent playing tennis and squash is 5 hours. As the time spent playing tennis increases, the time spent playing squash decreases, and we move along a straight line.

Part (b) shows a negative relationship such that as the journey length increases, the travel cost decreases as we move along a curve that becomes less steep.

Part (c) shows a negative relationship such that as leisure time increases, the number of problems worked decreases as we move along a curve that becomes steeper.

FIGURE A1.7 Maximum and Minimum Points



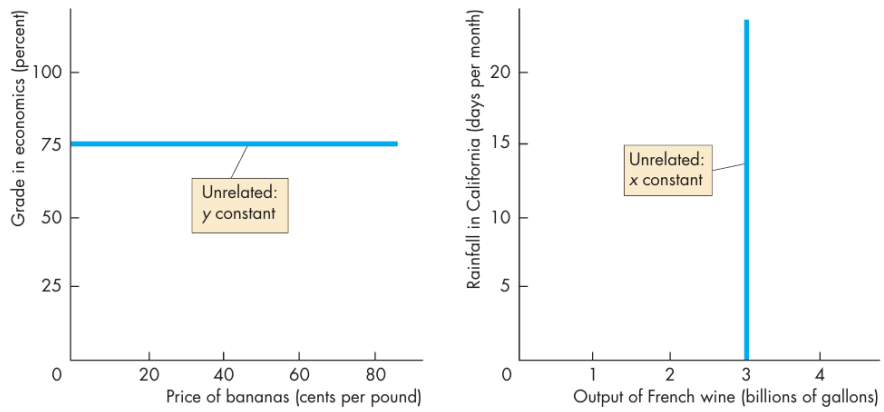
(a) Relationship with a maximum

(b) Relationship with a minimum

Part (a) shows a relationship that has a maximum point, *A*. The curve slopes upward as it rises to its maximum point, is flat at its maximum, and then slopes downward.

Part (b) shows a relationship with a minimum point, *B*. The curve slopes downward as it falls to its minimum, is flat at its minimum, and then slopes upward.

FIGURE A1.8 Variables That Are Unrelated



(a) Unrelated: *y* constant

(b) Unrelated: *x* constant

This figure shows how we can graph two variables that are unrelated. In part (a), a student's grade in economics is plotted at 75 percent on the *y*-axis regardless of the price of bananas on the *x*-axis. The curve is horizontal.

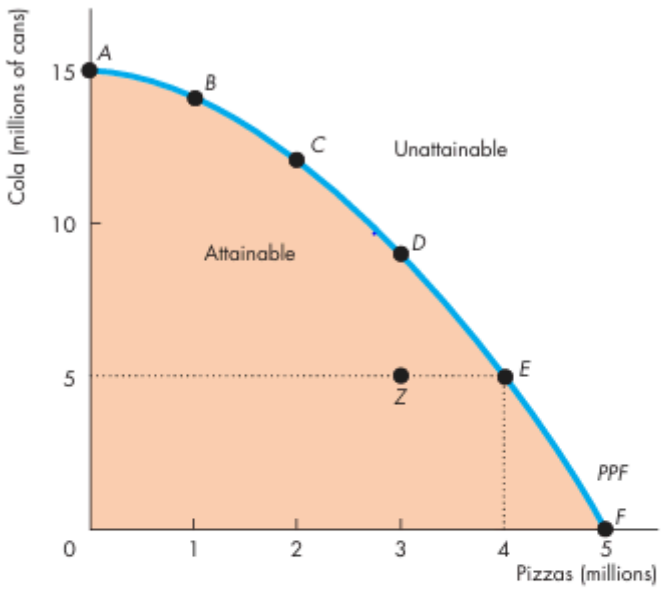
In part (b), the output of the vineyards of France on the *x*-axis does not vary with the rainfall in California on the *y*-axis. The curve is vertical.

PPF: Slide

The Production Possibilities Frontier (PPF)

- The PPF shows the maximum possible output combinations of two goods or services that an economy can produce using all resources efficiently.
- The production possibilities frontier (PPF) is the boundary between those combinations of goods and services that can be produced and those that cannot.
- Every working day, in mines, factories, shops, and offices and on farms and construction sites across the world people produce a vast variety of goods and services. But the quantities of goods and services that we can produce are limited by our available resources and by technology.

FIGURE 2.1 Production Possibilities Frontier



Possibility	Pizzas (millions)		Cola (millions of cans)
A	0	and	15
B	1	and	14
C	2	and	12
D	3	and	9
E	4	and	5
F	5	and	0

The production possibilities frontier for cola and pizza shows the limits to the production of these two goods, given the total resources and technology available to produce them.

The PPF illustrates scarcity because the points outside the frontier are unattainable. These points describe wants that can't be satisfied.

scribe wants that can't be satisfied. We can produce at any point inside the PPF or on the PPF. These points are attainable.

The table lists six production possibilities for cola and pizzas. Row A tells us that if we produce no pizzas, the maximum quantity of cola we can produce is 15 million cans. Points A, B, C, D, E, and F in the figure represent the rows of

the table. The curve passing through these points is the production possibilities frontier (PPF).
The PPF separates the attainable from the unattainable. Production is possible at any point inside the orange area or on the frontier. Points outside the frontier are unattainable. Points inside the frontier, such as point Z, are inefficient because resources are wasted or misallocated. At such points, it is possible to use the available resources to produce more of either or both goods.

Key concepts illustrated by the PPF:

- Scarcity: The frontier itself shows limits.
- Opportunity cost: Moving along the curve means giving up some of one good to gain more of another.
- Efficiency: Points on the PPF are efficient; points inside are inefficient; points outside are unattainable (without growth or trade).

Example: A country can produce either wheat or cloth. If it produces more wheat, it must reduce cloth production.

The **Production Possibilities Frontier (PPF)** shows the maximum possible combinations of two goods an economy can produce using all resources efficiently. For **example**, if a country can produce either **rice** or **cloth**, producing more rice means giving up some cloth due to limited resources. Suppose it can make 100 meters of cloth or 80 tons of rice; moving from cloth to rice production shows the **opportunity cost of the** amount of cloth forgone for each ton of rice produced. Points on the PPF represent full efficiency, points inside show underutilization, and points outside are unattainable without economic growth or better technology.

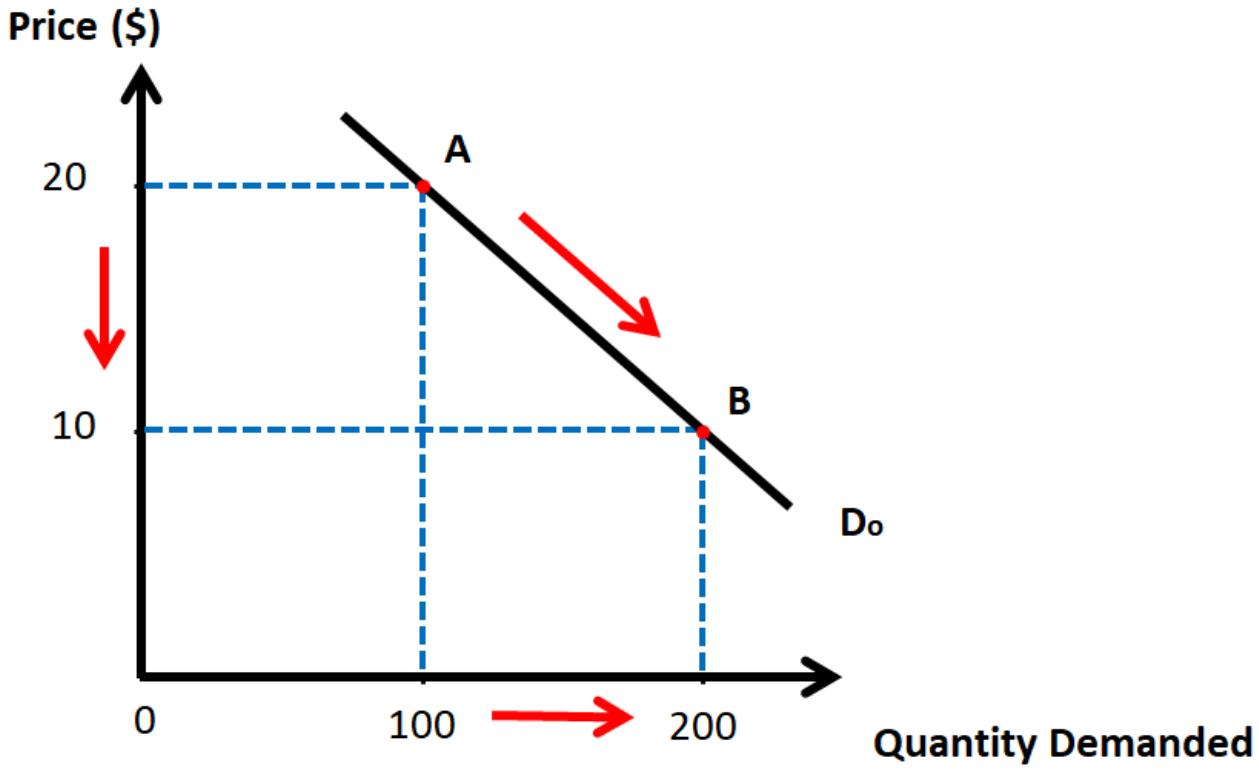
Opportunity Cost

- Opportunity cost is the value of the next best alternative forgone when a choice is made.
- Example: If a student spends time studying instead of working, the opportunity cost is the wage they could have earned. If Bangladesh invests more in garments, it may have fewer resources for agriculture.

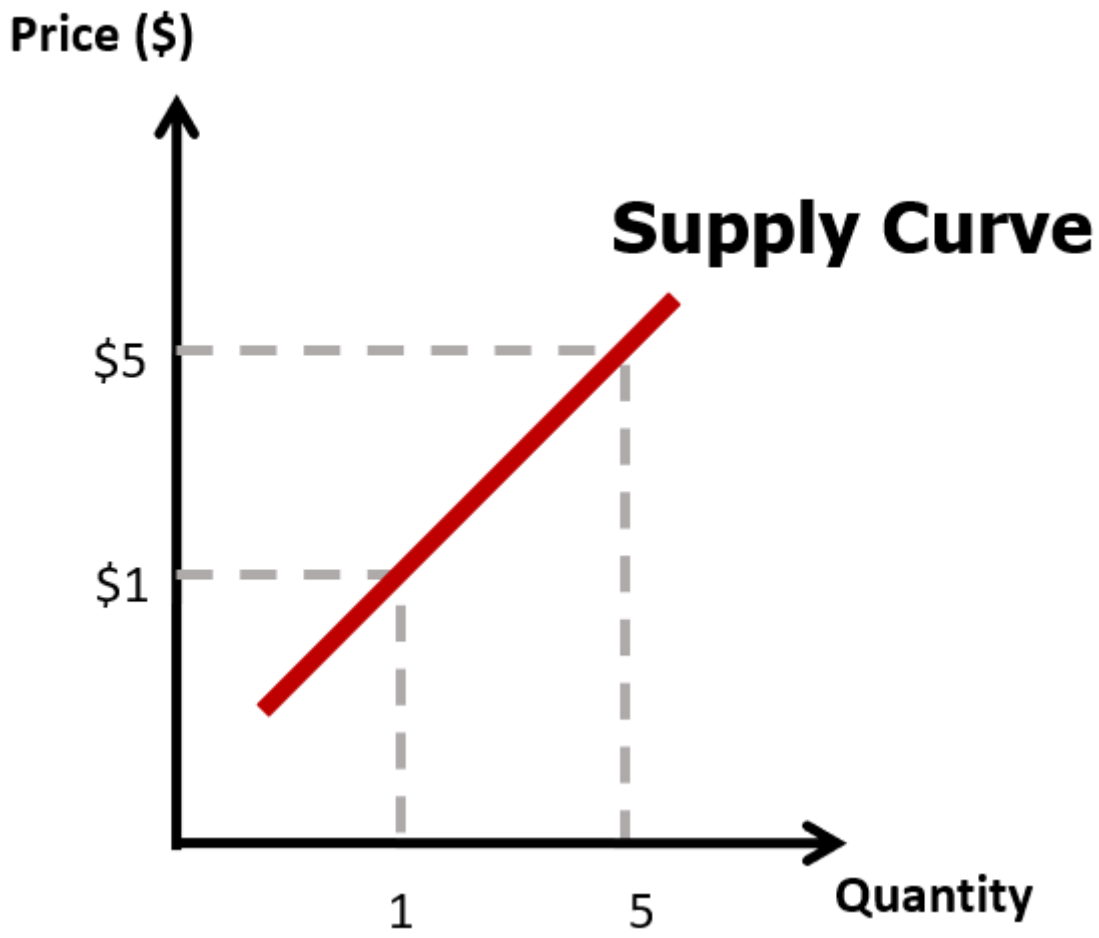
Basic Elements of Demand and Supply

a) Demand

- Law of Demand: As price falls, quantity demanded rises (ceteris paribus).
- Demand curve: Downward sloping, showing inverse relationship between price and quantity demanded.
- When onion imports are disrupted, demand stays high, but supply falls → prices rise sharply.



b) Supply



Forces behind the supply curve:

1. Production costs (input prices)
2. Technology improvements
3. Government policies (taxes, subsidies, regulations).
4. Number of sellers in the market
5. Expectations of future prices

Market Equilibrium

- Equilibrium occurs where the demand curve and supply curve intersect.
- At this point, quantity demanded = quantity supplied, and the market clears.
- Surplus (excess supply): When price is above equilibrium → downward pressure on price.
- Shortage (excess demand): When price is below equilibrium → upward pressure on price.

Efficiency in Markets

- When markets are competitive, equilibrium tends to be efficient:
    - Goods are produced at lowest cost (productive efficiency).
    - Resources are allocated to goods people value most (allocative efficiency).

However, market failures (like externalities, public goods, monopolies) may require government intervention.

Economics revolves around scarcity and choice. Every society must decide what, how, and for whom to produce. The PPF illustrates trade-offs and opportunity cost. Markets, command, and mixed economies provide different solutions to resource allocation. Demand and supply explain how markets work, with prices balancing buyers' and sellers' decisions. Efficiency is achieved when resources are used in the best possible way, though government intervention may be needed when markets fail.

Market Equilibrium

An equilibrium is a situation in which opposing forces balance each other. Equilibrium in a market occurs when the price balances buying plans and selling plans. The equilibrium price is the price at which the quantity demanded equals the quantity supplied. The equilibrium quantity is the quantity bought and sold at the equilibrium price.

If you demand something, then you
1. Want it.
2. Can afford it.
3. Plan to buy it.
Wants are the unlimited desires or wishes that people have for goods and services.

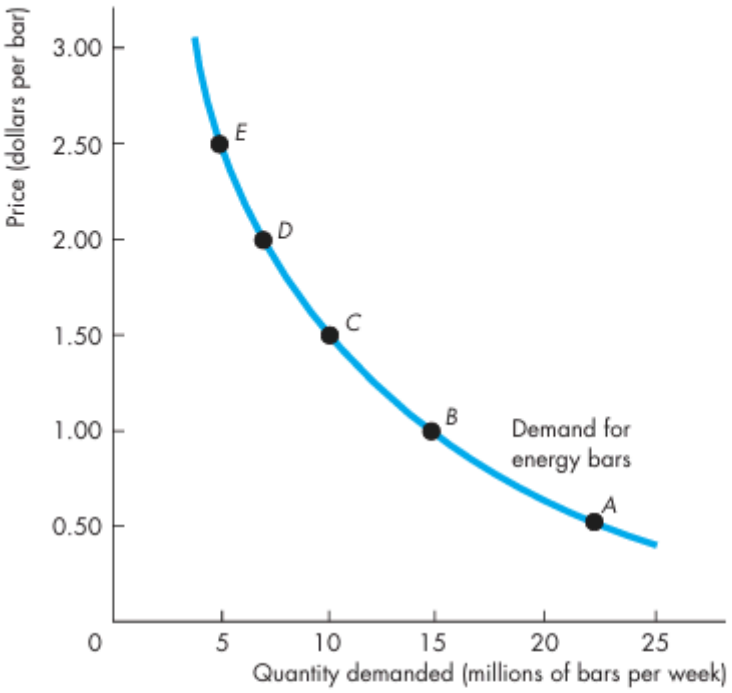
Demand curve and shifts of the demand curve

The law of demand states: Other things remain the same, the higher the price of a good, the smaller is the quantity demanded; and the lower the price of a good, the greater is the quantity demanded. Why does a higher price reduce the quantity demanded?
For two reasons:
- Substitution effect
- Income effect

Substitution Effect: When the price of a good rises, other things remain the same, its relative price, its opportunity cost rises. Although each good is unique, it substitutes other goods that can be used in its place. As the opportunity cost of a good rises, the incentive to economize on its use and switch to a substitute becomes stronger.
Income Effect: When a price rises, other things remain the same, the price rises relative to income. Faced with a higher price and an unchanged income, people cannot afford to buy all the things they previously bought. They must decrease the quantities demanded of at least some goods and services.

Normally, the goods whose price has increased will be one of the goods that people buy less of.
To see the substitution effect and the income effect at work, think about the effects of a change in the price of an energy bar. Several different goods are substitutes for an energy bar. For example, an energy drink could be consumed instead of an energy bar. Suppose that an energy bar initially sells for $3 and then its price falls to $1.50. People now substitute energy bars for energy drinks the substitution effect. And with a budget that now has some slack from the lower price of an energy bar, people buy even more energy bars for the income effect. The quantity of energy bars demanded increases for these two reasons. Now suppose that an energy bar initially sells for $3 and then the price doubles to $6. People now buy fewer energy bars and more energy drinks, the substitution effect. And faced with a tighter budget, people buy even fewer energy bars the income effect. The quantity of energy bars demanded decreases for these two reasons.

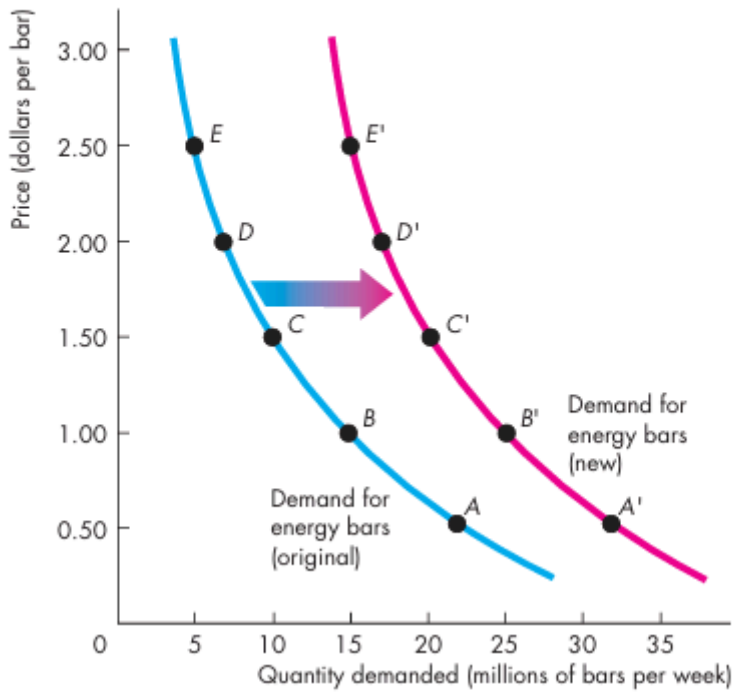
FIGURE 3.1 The Demand Curve



	Price (dollars per bar)	Quantity demanded (millions of bars per week)
A	0.50	22
B	1.00	15
C	1.50	10
D	2.00	7
E	2.50	5

The table shows a demand schedule for energy bars. At a price of 50¢ a bar, 22 million bars a week are demanded; at a price of $1.50 a bar, 10 million bars a week are demanded. The demand curve shows the relationship between quantity demanded and price, other things remain the same. The demand curve slopes downward: As the price falls, the quantity demanded increases.

FIGURE 3.2 An Increase in Demand



Original demand schedule Original income			New demand schedule New higher income		
	Price (dollars per bar)	Quantity demanded (millions of bars per week)		Price (dollars per bar)	Quantity demanded (millions of bars per week)
A	0.50	22	A'	0.50	32
B	1.00	15	B'	1.00	25
C	1.50	10	C'	1.50	20
D	2.00	7	D'	2.00	17
E	2.50	5	E'	2.50	15

A change in any influence on buying plans **other than the price** of the goods itself results in a new demand schedule and a shift of the demand curve. A change in income changes the demand for energy bars. At a price of $1.50 a bar, 10 million bars a week are demanded at the original income (row C of the table) and 20 million bars a week are demanded at the new higher income (row C'). A rise in income increases the demand for energy bars. The

demand curve shifts rightward, as shown by the shift arrow and the resulting red curve.

Six main factors bring changes in demand. They are changes in

- The prices of related goods
- Expected future prices
- Income
- Expected future income and credit
- Population
- Preferences

1. Changes in the Prices of Related Goods

A substitute is a good that can be used in place of another good.

A complement is a good that is used in conjunction with another good. Hamburgers and fries are complements, and so are energy bars and exercise. If the price of an hour at the gym falls, people buy more gym time and more energy bars.

Related goods are substitutes (used instead of each other) or complements (used together).

- Substitute goods: If the price of a substitute rises, demand for the other increases.
  Example (in the context of Bangladesh): If the price of beef rises, people may buy more chicken instead. As a result, the demand curve for chicken shifts rightward.

- Complementary goods: If the price of a complement rises, demand for the related good falls.
  Example: If the price of smartphones increases, fewer people may buy mobile internet packages. Thus, the demand curve for mobile data shifts leftward.

2. Changes in Expected Future Prices

When consumers expect that prices will rise in the future, they tend to buy more now; if they expect prices to fall, they delay purchases.
Example (Bangladesh): Before Ramadan, people expect food prices (like lentils or edible oil) to rise, so they buy earlier in bulk. This shifts the demand curve to the right temporarily.

3. Changes in Income

A **normal good** is one for which demand increases as income increases. An **inferior good** is one for which demand decreases as income increases. As incomes increase, the demand for air travel (a normal good) increases and the demand for long-distance bus trips (an inferior good) decreases.

People's income levels directly affect their ability to purchase goods and services.

- For normal goods, higher income increases demand.
  Example: As per capita income rises in Bangladesh, demand for branded clothing and air conditioners has increased rightward shift in the demand curve.

- For inferior goods, higher income reduces demand.
  Example: As people earn more, demand for low-quality rice or second-hand clothes decreases leftward shift in their demand curve.

4. Changes in Expected Future Income and Credit

If consumers expect higher future income or have easier access to loans, they are more confident to spend now.

Example (Bangladesh): When salaried workers expect a government pay scale revision, they often increase current spending on home appliances or electronics. This raises present demand, shifting the demand curve rightward.

a salesperson gets the news that she will receive a big bonus at the end of the year, so she goes into debt and buys a new car right now, rather than waiting until she receives the bonus.

5. Changes in Population

The size and structure of the population also affect demand.
Example: Bangladesh's growing urban population increases demand for apartments, ride-sharing services, and fast food. As population rises, the demand curve for these goods and services shifts rightward.
Conversely, if migration or demographic shifts reduce the number of consumers, demand may decrease—leftward shift.

For example, the demand for parking spaces, running shoes, movies, or just about anything that you can imagine is much greater in New York City (population 8.3 million) than it is in Boise, Idaho (population 212,000).

6. Changes in Preferences

Demand depends on preferences. Preferences determine the value that people place on each good and service. Preferences depend on such things as the weather, information, and fashion. For example, greater health and fitness awareness has shifted preferences in favor of energy bars, so the demand for energy bars has increased.

Tastes, fashion, and social attitudes influence demand.
Example (Bangladesh): With growing health awareness, people prefer low-fat dairy products and organic foods. So, demand for organic vegetables shifts rightward, while demand for sugary drinks shifts leftward.
Similarly, social trends or advertising (like popularity of smartphones or e-commerce) can change preferences and shift demand.

Summary

A shift in the demand curve means that at every price level, people are willing to buy more or less of a good.

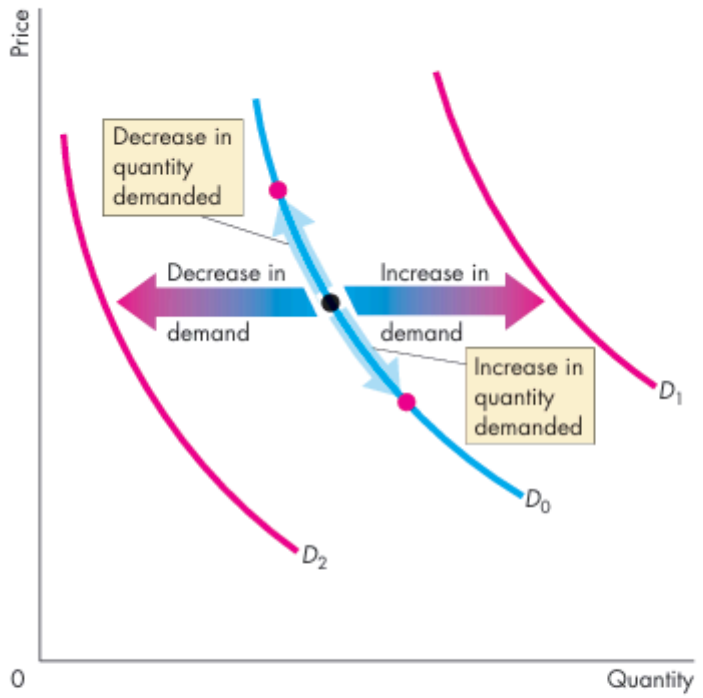
- Rightward shift → Increase in demand
- Leftward shift → Decrease in demand

In Bangladesh's economy, such shifts occur constantly due to changes in income, lifestyle, and market expectations—reflecting a dynamic and growing consumer market.

Movement Along the Demand Curve If the price of the good changes but no other influence on buying plans changes, we illustrate the effect as a movement along the demand curve.

A Shift of the Demand Curve If the price of a good remains constant but some other influence on buying plans changes, there is a change in demand for that good. We illustrate a change in demand as a shift of the demand curve. For example, if more people work out at the gym, consumers buy more energy bars regardless of the price of a bar. That is what a right ward shift of the demand curve shows—more energy bars are demanded at each price.

FIGURE 3.3 A Change in the Quantity Demanded Versus a Change in Demand



When the price of the good changes, there is a movement along the demand curve and *a change in the quantity demanded*, shown by the blue arrows on demand curve D_0. When any other influence on buying plans changes, there is a shift of the demand curve and a *change in demand*. An increase in demand shifts the demand curve rightward (from D_0 to D_1). A decrease in demand shifts the demand curve leftward (from D_0 to D_2).

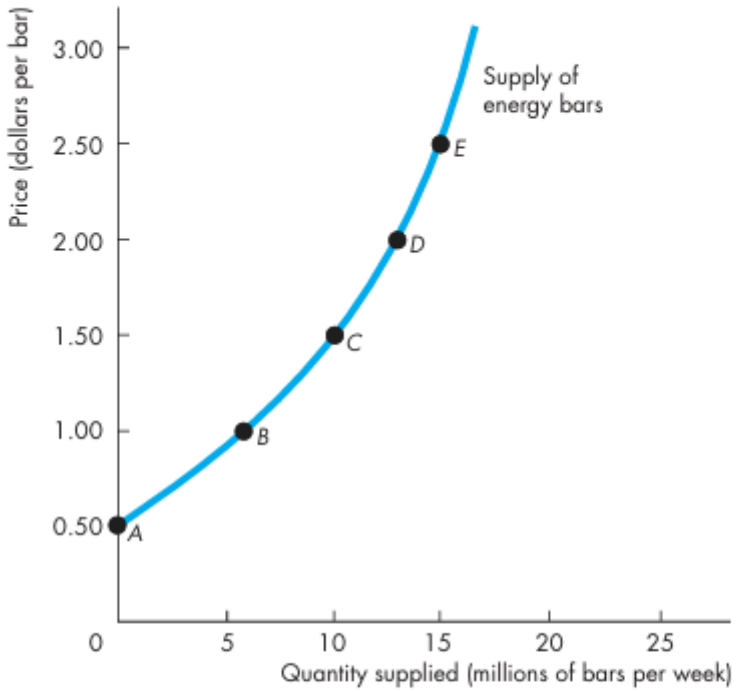
The marginal cost of a good is the opportunity cost of producing one more unit of it.

If a firm supplies a good or service, the firm 1. Has the resources and technology to produce it. 2. Can profit from producing it. 3. Plans to produce it and sell it. A supply is more than just having the resources and the technology to produce something. Resources and technology are the constraints that limit what is possible. Many useful things can be produced, but they are not produced unless it is profitable to do so. Supply reflects a decision about which technologically feasible items to produce.

The law of supply states: Other things remaining the same, the higher the price of a good, the greater is the quantity supplied; and the lower the price of a good, the smaller is the quantity supplied

Why does a higher price increase the quantity supplied? It is because marginal cost increases. As the quantity produced of any good increases, the marginal cost of producing the good increases.

FIGURE 3.4 The Supply Curve



	Price (dollars per bar)	Quantity supplied (millions of bars per week)
A	0.50	0
B	1.00	6
C	1.50	10
D	2.00	13
E	2.50	15

1. Changes in the Prices of Factors of Production

Factors of production include land, labor, capital, and raw materials. When their prices change, production costs change.

- If production costs rise, producers can supply less at each price → Supply curve shifts left.

- If costs fall, producers can supply more → Supply curve shifts right.

Example (Bangladesh):
If the price of diesel fuel increases, the cost of running irrigation pumps and transport for farmers rises. As a result, the supply of rice decreases (leftward shift).
On the other hand, if fertilizer prices fall due to government subsidies, farmers can produce more rice (rightward shift).

2. Changes in the Prices of Related Goods Produced

Sometimes, a firm can produce several goods using the same resources. The decision to produce one product depends on the profitability of others.

- If the price of one product increases, producers may switch resources to make more of that, reducing supply of the other → Leftward shift.

- If the price of one related product falls, producers may produce more of the other → Rightward shift.

Example (Bangladesh):
A textile factory can produce either T-shirts or jeans. If the price of jeans rises in the export market, the factory may produce more jeans and fewer T-shirts. This reduces the supply of T-shirts (leftward shift).

3. Changes in Expected Future Prices

If producers expect that the price of their product will rise in the future, they might withhold supply now to sell later.

- Expectation of higher future prices → Current supply decreases → Leftward shift.

- Expectation of lower future prices → Current supply increases → Rightward shift.

Example (Bangladesh):
If jute exporters expect international jute prices to increase, they may store their products instead of selling immediately, causing a temporary decrease in current supply (leftward shift).

4. Changes in the Number of Suppliers

More firms entering the market increases supply; fewer firms decrease supply.

- Increase in number of suppliers → Rightward shift.

- Decrease in number of suppliers → Leftward shift.

Example (Bangladesh):
With more new poultry farms opening, the supply of eggs and chicken increases (rightward shift).
If small farms close down due to high feed costs, the overall supply decreases (leftward shift).

5. Changes in Technology

Technological improvement makes production more efficient, lowering costs and increasing supply.

- Better technology → Rightward shift.

- Outdated or damaged technology → Leftward shift.

Example (Bangladesh):
The introduction of modern sewing machines and automated cutting in the garment industry increases output- rightward shift in supply.

If factories face machinery breakdowns or power outages, production drops — leftward shift.

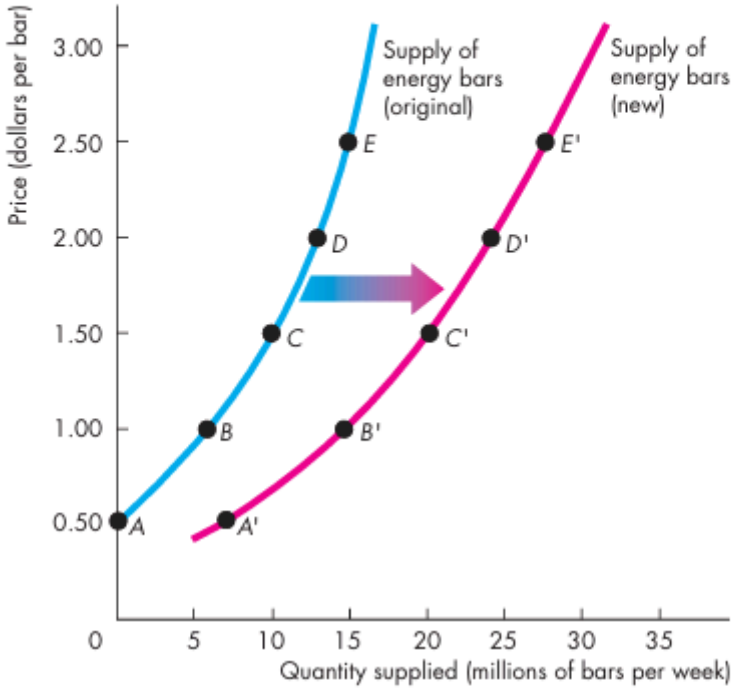
6. The State of Nature

Natural conditions like weather, floods, droughts, and other disasters directly affect production, especially in agriculture.

- Favorable conditions → Rightward shift.
- Unfavorable conditions → Leftward shift.

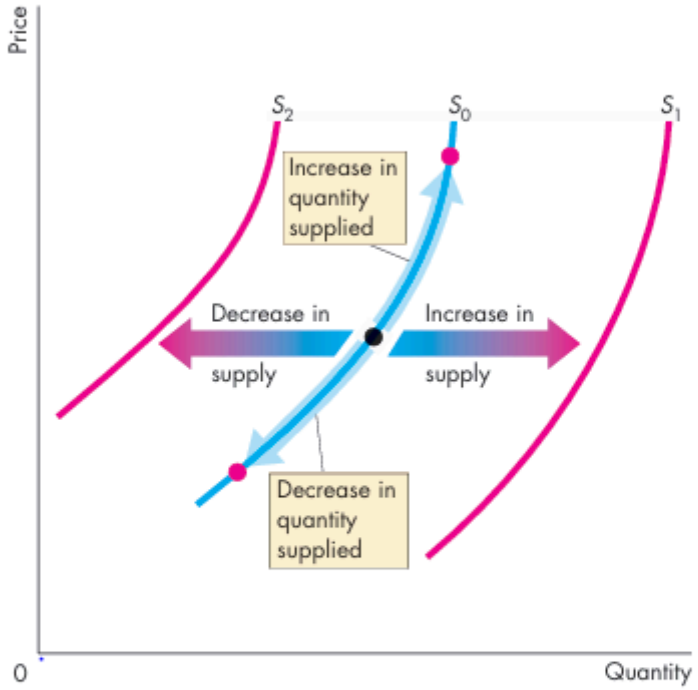
Example (Bangladesh):
Good monsoon rainfall increases rice production, shifting supply rightward.
In contrast, floods or cyclones damaging crops in coastal areas reduce supply (leftward shift).

FIGURE 3.5 An Increase in Supply



Original supply schedule Old technology			New supply schedule New technology		
	Price (dollars per bar)	Quantity supplied (millions of bars per week)		Price (dollars per bar)	Quantity supplied (millions of bars per week)
A	0.50	0	*A'*	0.50	7
B	1.00	6	*B'*	1.00	15
C	1.50	10	*C'*	1.50	20
D	2.00	13	*D'*	2.00	25
E	2.50	15	*E'*	2.50	27

FIGURE 3.6 A Change in the Quantity Supplied Versus a Change in Supply



When the price of the good changes, there is a movement along the supply curve and a *change in the quantity supplied*, shown by the blue arrows on supply curve S_0. When any other influence on selling plans changes, there is a shift of the supply curve and a *change in supply*. An increase in supply shifts the supply curve rightward (from S_0 to S_1), and a decrease in supply shifts the supply curve leftward (from S_0 to S_2).

We have seen that when the price of a good rises, the quantity demanded decreases and the quantity supplied increases. We are now going to see how the price adjusts to coordinate buying plans and selling plans and achieve an equilibrium in the market. An equilibrium is a situation in which opposing forces balance each other. Equilibrium in a market occurs when the price balances buying plans and selling plans. The equilibrium price is the price at which the quantity demanded equals the quantity supplied. The equilibrium quantity is the quantity bought and sold at the equilibrium price. A market moves toward its equilibrium because ■ Price regulates buying and selling plans. ■ Price adjusts when plans don't match.

Price as a Regulator The price of a good regulates the quantities demanded and supplied. If the price is too high, the quantity supplied exceeds the quantity demanded. If the price is too low, the quantity demanded exceeds the quantity supplied. There is one price at which the quantity demanded equals the quantity supplied. Let's work out what that price is. Figure 3.7 shows the market for energy bars. The table shows the demand schedule (from Fig. 3.1) and the supply schedule (from Fig. 3.4). If the price is 50¢ a bar, the quantity demanded is 22 million bars a week but no bars are supplied. There is a shortage of 22 million bars a week. The final column of the table shows this shortage. At a price of $1.00 a bar, there is still a shortage but only 9 million bars a week. If the price is $2.50 a bar, the quantity supplied is 15 million bars a week but the quantity demanded is only 5 million. There is a surplus of 10 million bars a week. The one price at which there is neither a shortage nor a surplus is $1.50 a bar. At that price, the quantity demanded equals the quantity supplied: 10 mil lion bars a week. The equilibrium price is $1.50 a bar, and the equilibrium quantity is 10 million bars a week. Figure 3.7 shows that the demand curve and the supply curve intersect at the equilibrium price of $1.50 a bar. At each price above $1.50 a bar, there is a surplus of bars. For example, at $2.00 a bar, the surplus is 6

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